

S.A. ENGINEERING COLLEGE

Autonomous Institution | Affiliated to Anna University | NBA & NAAC 'A' Grade

Dept of Electronics & Communication Engg & Dept of Electronics Engg [VLSI Design & Technology]

KERNEL PRIME '26 - SOFTWARE TRACK

National Level 24-Hour Hackathon | October 8-9, 2026

OFFICIAL PROBLEM STATEMENT: PS-08

Privacy-Preserving Zero-Knowledge Digital Identity Vault

Domain: Cryptography / Identity / Privacy Tech

1. PROBLEM OVERVIEW & CHALLENGE STATEMENT

Implement a selective disclosure decentralized identity framework utilizing zk-SNARKs to allow citizens to prove age, citizenship, and academic credentials without revealing underlying personal data.

2. MANDATORY ARCHITECTURAL REQUIREMENTS

[REQ-1] zk-SNARK / Groth16 circuit implementation for private predicate verification (e.g., Age ≥ 18).

[REQ-2] W3C Verifiable Credentials and Decentralized Identifiers (DID) standard compliance.

[REQ-3] Hardware secure enclave / biometric key generation on client devices.

[REQ-4] Instant revocation check mechanism using dynamic cryptographic accumulator trees.

[REQ-5] Seamless single-click OAuth2 / OIDC bridge for rapid web application integration.

3. HACKATHON EVALUATION MATRIX

* System Architecture & Concurrency Resilience: 25%

* Technical Innovation & Production Viability: 25%

* Live Demo & Real-World Latency Benchmarks: 30%

* Security Hardening, UI Polish & Documentation: 20%

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